



Inferring local to regional changes in forest composition from Holocene macrofossils and pollen of a small lake in central Upper Michigan

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ABSTRACT

Vegetational response to climatic change involves processes of population and community dynamics within local stands, which scale up to landscape-level changes in vegetation composition and broad-scale changes in species distributions. Understanding these dynamics poses a critical challenge to paleoecological explanation, because of the broad range of scales at which these dynamics take place and interact. We present an 8600-year paleoecological record of local and regional changes in forest composition from a small (2.6 ha) lake in central Upper Michigan. Plant macrofossils provide a spatially precise record of local forest composition, while pollen data provide a spatially integrated record of regional vegetational changes. Temporal patterns among different macrofossil types within species show overall coherence, indicating that changes in macrofossil abundance generally record changes in local tree abundance. Temporal patterns in macrofossil sequences correspond to patterns in pollen sequences, indicating that local changes contributed to the large-scale changes in the surrounding region. The pollen and macrofossil records show nearly continuous turnover in vegetation composition throughout the past 8600 years; the longest period without major compositional change was ca 1600 years, and dynamics at multidecadal to multicentennial scales are observed during many periods. Coordinated application of temporally precise sequences of pollen and macrofossil data at multiple sites can support inferences concerning vegetation dynamics at multiple spatial and temporal scales, and test mechanistic hypotheses.

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1. Introduction

A central goal of paleoecology is to infer properties and dynamics of past individuals, populations, communities, and ecosystems from multiple lines of fragmentary, incomplete evidence. In terrestrial plant paleoecology, pollen analysis has been employed to these ends for nearly a century (Von Post, 1916), and Quaternary plant macrofossil analysis is long past its sesquicentennial (Steenstrup, 1841; Vaupell, 1851). Quaternary pollen records are

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legion across much of the terrestrial globe (Williams et al., 2004; Bartlein et al., 2011; Brewer et al., 2012), and plant macrofossil records, though sparser, number in the hundreds in some regions (e.g., Betancourt et al., 1990; Jackson et al., 1997; Binney et al., 2009). Ecological applications of pollen and macrofossil records rest on a foundation of empirical studies of taphonomic processes, comparative studies of modern assemblages and vegetation, and formal and conceptual models for correcting biases and facilitating interpretation.

In spite of the widespread application of pollen and plant macrofossil data and the considerable body of knowledge underlying their interpretation, substantial uncertainties remain. Pollen assemblages represent distance-weighted integrations of vegetation composition extending from the immediate vicinity of the

depositional basin out to tens of kilometers away, with distance-weightings varying among taxa according to their pollen-dispersal properties (Prentice, 1985; Sugita, 1994; Jackson and Lyford, 1999). By itself, a pollen assemblage corresponds to no single entity that might otherwise be of interest to ecologists; its primary interest lies in its having recorded features of vegetation that can no longer be observed directly. In aggregating innumerable individuals, populations, and patches over a broad landscape, a pollen assemblage conveys a sense of 'average' composition of regional vegetation. However, the regional signal may be distorted by pollen from vegetation near the depositional site, which may differ substantially from the regional average (Bradshaw and Webb, 1985; Jackson, 1990, 1994).

Differentiating between pollen deriving from local and distant sources has long been a central challenge in pollen analysis; insufficient ability to do so is a major source of uncertainty in ecological and biogeographic applications (Davis et al., 1991; MacDonald, 1993; McLachlan and Clark, 2004). A few tools, based on empirical and theoretical understanding of pollen dispersal, are available to support such differentiation. The proportion of local pollen in sediments is inversely (and non-linearly) related to the size of the depositional basin (Prentice, 1985; Jackson, 1994; Sugita, 1994, 2007a, 2007b). Thus, site selection can dampen or amplify local representation, depending on the purpose of the study (Jacobson and Bradshaw, 1981). Application of pollen-dispersal models to spatial networks of small and large basins can support identification of local and regional sources and detect spatial patchiness of vegetation (Sugita, 1994, 2007a, 2007b). Initial applications are promising (Soepboer et al., 2010; Sugita et al., 2010), but are necessarily restricted to regions rich in depositional basins of suitable size.

A complementary approach is paired analysis of pollen and plant macrofossils. In contrast to pollen, macrofossils from lake sediments are indisputably local in origin; airborne dispersal is, except in rare cases, limited to 10^0 – 10^2 m, and waterborne macrofossils can derive from no further than the catchment boundary. Thus, a macrofossil assemblage derives from a relatively small area surrounding the lake, more analogous to the scale of conventional ecological observation and sampling than that of a pollen assemblage. Macrofossil assemblages, however, are subject to their own biases of representation, and vegetation adjacent to a lakeshore may not be representative of surrounding uplands. Furthermore, sample sizes for macrofossil assemblages tend to be small, owing to low concentrations in sediments (typically at least five orders of magnitude lower than pollen concentrations) and to the low sediment volume obtained in typical lake-sediment cores.

Macrofossil data tend to be treated conservatively in paleoecological inference. In the canonical text of pollen analysis, Knut Fægri asserted that although occurrence of a macrofossil might indicate local presence at a site, "nothing can be concluded from its absence" (Fægri and Iversen, 1975, p. 123). Accordingly, many studies use macrofossil data exclusively as singular indicators of local occurrence. This is reasonable when sample sizes are low, taphonomic biases are high, or sampling is haphazard or opportunistic. However, local presence and absence can be reliably inferred from macrofossil data, provided sample sizes are large enough and taphonomic biases are such that they maximize likelihood of representation in sediments given occurrence in surrounding vegetation (Jackson, 2012). Presence/absence at individual sites can be scaled up to infer past geographic distributions of species (Jackson et al., 1997, 2000; Binney et al., 2009).

Macrofossils are often tabulated and expressed as concentrations (number per unit sediment volume), accumulation rates (numbers per cm^2 per year), or percentages of a designated macrofossil sum. Quantitative macrofossil sequences are often

based on small sample sizes and have high inter-sample variability, making it difficult to differentiate between vegetational signal and extraneous noise. Nonetheless, macrofossil and pollen data often show parallel patterns (e.g., Jackson and Booth, 2007), suggesting that macrofossil data can support inferences concerning local-scale vegetation composition and dynamics. Macrofossil data may be underutilized as a source of information concerning vegetational change at local and regional scales, and as a tool for linking the regional changes represented in pollen sequences to the local changes that might be observed at the stand scale.

Effective study design and robust interpretation of macrofossil data can be informed by principles and observations based on empirical studies and theoretical considerations. Because macrofossil sources are largely limited to trees growing within 10^0 – 10^2 m of shore, airborne macrofossil flux-density to a lake surface will increase as an approximate linear function of shoreline perimeter, but total lake area will increase as an approximate square function of the perimeter. Therefore, all else held constant, macrofossil flux density per unit lake area will decrease rapidly as lake size increases. Macrofossil density in benthic sediments decreases with distance from shore (Wainman and Mathewes, 1990; Jackson and Booth, 2007), owing to attenuating airborne dispersal to the lake surface and to diffusion as airborne, floating, or suspended macrofossils move further from onshore sources. As a general rule, macrofossil preservation in sediments increases with water depth (Jackson, 1989; Jackson and Booth, 2007); lower temperatures and oxygen deficits retard decomposition. Emergent or-floating-leaved vegetation in shallow waters pose physical barriers to dispersal of macrofossils to deeper waters, and will thus decrease macrofossil concentrations in offshore sediments. Steep slopes along a lake-shore will increase adjacency of upland trees to open water, and overhanging trees will contribute macrofossils directly to the water surface while inhibiting emergent vegetation by shading the narrow littoral margin of the lake. Accordingly, small lakes with deep water within 50 m of shore bordered by steep, forested slopes and overhanging trees are ideally suited to yield high concentrations of well-preserved macrofossils.

In this paper, we compare Holocene pollen and macrofossil data, obtained at relatively high (decades to centuries) temporal resolution, from sediments of Tower Lake, a small lake in northern Michigan, specifically selected to maximize representation of plant macrofossils from forest species characteristic of the surrounding upland landscape. We compare the pollen and macrofossil data for 11 tree species, assessing the extent to which patterns in the macrofossil record correspond to those in the pollen record. This comparison supports assessment of the local forest dynamics represented in the macrofossil record in terms of the regional dynamics recorded by pollen. We also compare temporal patterns among different macrofossil types within tree taxa, to determine whether those patterns are attributable to changes in local tree populations or to taphonomic or other factors. We discuss the various factors, taphonomic and other, that can influence macrofossil abundance in lake sediments, and suggest how paleoecological studies can be effectively designed to maximize and differentiate information on both local and regional changes in forest composition. Finally, we discuss the Tower Lake record in terms of the challenges faced by paleoecologists in explaining ecological dynamics that involve multiple temporal and spatial scales, and how these challenges might be addressed by careful consideration of scale and employment of multiple lines of evidence.

2. Site description

Tower Lake (informal name) is an endorheic kettle lake in central Upper Michigan ($46^\circ 32' 31''\text{N}$, $86^\circ 02' 13''\text{W}$, 261 m elevation)

(Fig. 1). We selected the site to maximize representation of plant macrofossils in sediments and also to maximize representation of pollen from relatively local sources. Key attributes were the lake's small size (2.6 ha), adjacent upland slopes with overhanging trees, deep water (12.0 m maximum depth), and absence of a littoral zone dominated by emergent aquatic plants.

Tower Lake is surrounded by second-growth forests dominated by *Fagus grandifolia*, *Tsuga canadensis*, *Acer saccharum*, *Betula alleghaniensis*, and *Pinus strobus*; individuals of all five species grow within 20 m of the lakeshore adjacent to our coring site. Other forest constituents near the lake include *Quercus rubra*, *Fraxinus americana*, *Picea glauca*, *Betula papyrifera*, *Abies balsamea*, and *Prunus serotina*. The surrounding region is mapped as “beech-sugar maple–hemlock forest” based on early 19th Century General Land Office surveys (Albert and Comer, 2008). These forests occupied glacial till, and comprised elements of an edaphically controlled mosaic, with *P. strobus*–*Pinus resinosa* forests on well-drained outwash plains and *Picea*–*Abies* conifer swamps in poorly drained sites. *Pinus*-dominated outwash plains are scarce within 10 km of Tower Lake, though extensive areas occur some 15 km both east and west of the lake.

Tower Lake is in coarse-textured till on an end moraine (Farrand and Bell, 1982). The low-resolution State Soil Geographic database (STATSGO) maps soils near Tower Lake as deep, well-drained sandy

soils with ortsteins (cemented Al and Fe oxides in the B horizon). Direct inspection within the lake catchment confirms that soils are spodosols, formed on substrates of sand, loamy sand, and sandy loam, with a moderately to extremely well-cemented ortstein horizon 25–70 cm below the ground surface.

Water levels at Tower Lake are relatively stable, with seasonal and interannual fluctuations on the order of <50 cm (based on shoreline observations). Scattered populations of *Nuphar*, *Potamogeton epiphydrus*, *Sparganium angustifolium*, *Dulichium arundiacum*, and *Equisetum* cf. *fluviatile* grow in shallow water near the lakeshore. A floating *Sphagnum* mat occurs on the west side of the lake, opposite our coring area, and a small mat occupies an embayment on the northeast side (Fig. 1).

3. Methods

We obtained sediment cores at water depths of 10–11 m, at sites 35–45 m from the eastern margin of the lake, using a Wright–Cushing modified Livingstone piston corer (7.5 cm diameter). Upper sediments were highly flocculent, and on our first coring trip (2004; Core 1) we could not recover the uppermost meter owing to failure of an extruding device. We obtained and extruded a top-meter core (Core 4) in 2005, but radiocarbon dates indicated a temporal gap between its base and the uppermost sediments of Core 1. Additional cores (6 & 7) were obtained in 2006 to cover the gap (Appendix A).

Age models were based on 17 accelerator mass spectrometry (AMS) radiocarbon dates for terrestrial plant macrofossils (Table 1). Dates were converted to calendar years using the INTCAL calibration curve (Reimer et al., 2009). Age models were developed using the Bayesian method of Blaauw and Christen (2011), using empirical estimates of sediment-accumulation rates and temporal autocorrelation from similar sites as priors (Goring et al., 2012). We developed a separate age-model for each of the four cores (Fig. 2), and developed a composite pollen and macrofossil profile based on the four temporally overlapping cores (Fig. 2, Appendix A). All ages are expressed in calibrated calendar years before present. Pollen and macrofossil profiles for individual cores are provided in Appendix B.

Pollen samples comprised 1 cm³, representing 1 cm of sediment depth. Each macrofossil sample also represented a 1-cm sediment interval, with sample volumes ranging from 15 to 30 cm³. In general, each 1-cm increment represented between 10 and 50 years' elapsed time. We initially obtained 1-cm thick core segments at 5 cm intervals throughout the cores for macrofossil analysis. We increased sampling density in the sediment column at selected intervals to pinpoint the first occurrence of *B. alleghaniensis* macrofossils. After those initial analyses, we obtained contiguous samples, 1 cm in thickness, at least 2 cm above and 5 cm below the first macrofossil occurrence of that species. Finally, we analyzed contiguous macrofossil samples, each 1 cm thick, from sediments dating between approximately 6100 and 4600 yr BP. Each sediment sample for macrofossil analysis was measured volumetrically by water displacement, dispersed in water, and sieved (710 µm mesh). Macrofossils were scanned under a stereomicroscope, identified, and tallied, and all macrofossil counts were standardized to a sample volume of 25 cm³.

Plant macrofossils were identified using a modern reference collection. Needles of *P. resinosa* and *Pinus banksiana* were differentiated based on number and position of resin ducts in cross-section (Harlow, 1947). Selected bud scales were identified to species based on comparison with modern reference specimens, using the following criteria: *P. strobus* bud scales are elongated, usually inrolled, and moderately sclerified. *A. saccharum* bud scales are highly sclerified, with pointed tips, and dense trichomes on the

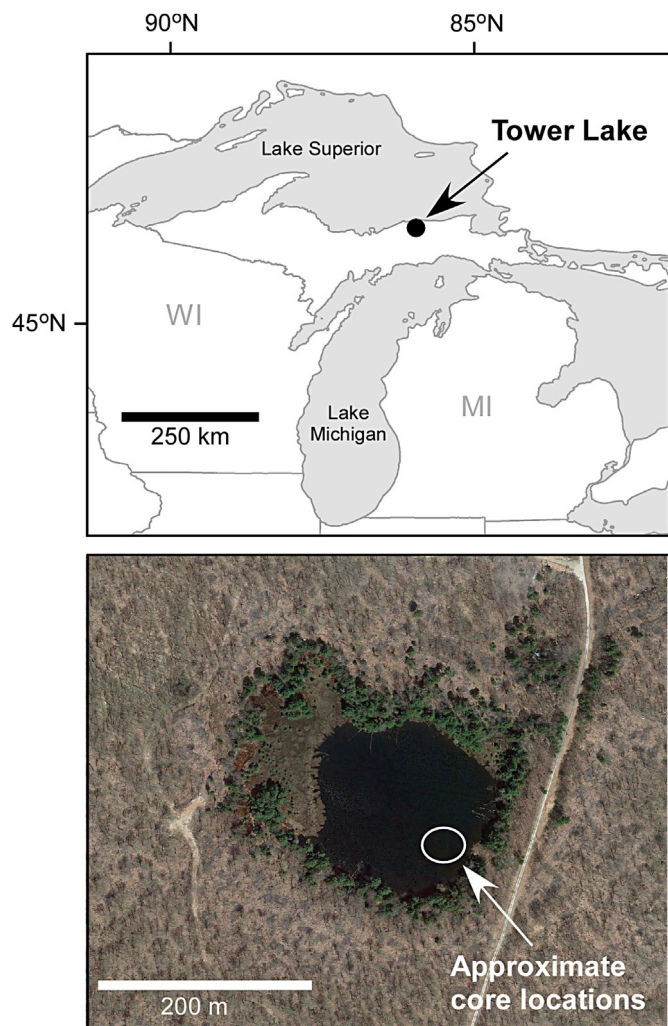


Fig. 1. Location of Tower Lake in the western Great Lakes region of North America (upper) and of the coring sites (white circle) within Tower Lake (lower).

Table 1
AMS radiocarbon dates from Tower Lake.

Core	Depth (cm)	Material dated	Laboratory number	¹⁴ C yr BP	Calibrated age (yr BP) ^a
6	58–59	<i>Betula alleghaniensis</i> fruits, <i>Fagus grandifolia</i> bud scales, <i>Acer saccharum</i> bud scales, <i>Tsuga canadensis</i> seeds, <i>Pinus strobus</i> needle	UGAMS-01321	398 ± 43	454 (316–518)
4	59–60	<i>Betula alleghaniensis</i> fruits, <i>Pinus strobus</i> needles	UGAMS-01207	464 ± 41	513 (335–617)
7	86–87	<i>Betula alleghaniensis</i> fruits and bracts, <i>Acer saccharum</i> budscale, <i>Tsuga canadensis</i> needles and microstrobili, <i>Pinus strobus</i> bud scales, needles and microstrobili	UGAMS-01322	965 ± 55	862 (743–964)
7	142–143	<i>Betula alleghaniensis</i> fruits and bract, <i>Tsuga canadensis</i> seed and needles, and <i>Pinus strobus</i> needles and bud scales	UGAMS-01601	1772 ± 43	1691 (1569–1817)
6	147–148	<i>Betula alleghaniensis</i> fruits and bracts, <i>Tsuga canadensis</i> seed and needles, <i>Pinus strobus</i> needles and bud scales	UGAMS-01320	1796 ± 40	1728 (1607–1856)
7	176–177	<i>Betula alleghaniensis</i> fruits and bract, <i>Tsuga canadensis</i> cone scale	UGAMS-01323	2220 ± 43	2232 (2140–2339)
1	90–91	<i>Acer saccharum</i> seed	UGA-15423	2400 ± 50	2450 (2340–2701)
1	155–156	<i>Betula alleghaniensis</i> fruits and bracts	AA-73830	3409 ± 37	3660 (3564–3823)
1	204–205	<i>Pinus strobus</i> needles	UGA-15424	3580 ± 50	3883 (3719–4068)
1	239–240	<i>Pinus strobus</i> needles	NOSAMS-62963	4160 ± 30	4705 (4581–4827)
1	256–257	<i>Pinus strobus</i> needles and seed	UGAMS-03084	4290 ± 30	4853 (4826–4959)
1	260–261	<i>Betula alleghaniensis</i> fruit, <i>Betula</i> sp. fruit, <i>Pinus strobus</i> needles and bud scales, <i>Tsuga</i> needles, twigs, and microstrobili	UGAMS-03074	4480 ± 30	5173 (4979–5008)
1	275–276	<i>Tsuga</i> needles	NOSAMS-62962	4750 ± 35	5513 (5329–5586)
1	281–282	<i>Betula papyrifera</i> bract, <i>Tsuga canadensis</i> seeds, <i>Pinus strobus</i> needles	UGAMS-01209	4701 ± 48	5425 (5319–5581)
1	336–337	<i>Pinus strobus</i> needles	UGAMS-01210	5594 ± 49	6370 (6294–6458)
1	390–391	<i>Pinus strobus</i> bud scales and microstrobili	AA-73828	6756 ± 43	7614 (7524–7677)
1	443–444	<i>Betula papyrifera</i> fruits and bracts	UGAMS-01051	7904 ± 55	8740 (8594–8979)

^a Age estimates include the median intercept and the minimum and maximum ages in parentheses based on 2 standard deviations from minimum and maximum intercepts.

inner scale surface at the apex. *Q. rubra* bud scales have a highly sclerified opaque apex and a moderately sclerified, translucent basal portion. *F. grandifolia* bud scales are moderately translucent, but most have a thickened, opaque band. They are generally longer than *Quercus* bud scales and have sparse to moderately dense trichomes. Some bud-scale morphotypes were not identified or tallied. Macrofossil analyses and identifications were carried out by Reeves and Jackson. Additional samples to increase temporal density were analyzed by Andersen and Jackson.

Pollen samples were prepared using standard chemical digestion techniques for unconsolidated lake sediments (Fægri and Iversen, 1975; Jackson, 1999). Individual pollen samples were counted to a minimum sum of 250 arboreal grains; samples were counted by Reeves, Minckley, Andersen, and Jones (details in Appendix C). Pollen percentages were calculated using an arboreal pollen sum, a total pollen sum for all other pollen taxa, and a total pollen and spore sum for spore taxa. Pollen zones were demarcated with the aid of CONISS, a stratigraphically constrained cluster analysis (Grimm, 1987). We have already published a pollen sequence, sampled at contiguous 1-cm intervals, from sediments spanning 6100–4600 yr BP at Tower Lake (Booth et al., 2012a). We use that data set in this paper. However, that data set included substantial errors in subgeneric differentiation of *Pinus* pollen; all counts with subgeneric *Pinus* differentiations used in our pollen diagram are from other data sets counted by other analysts (see Appendix C for details). Samples from the contiguous series 6100–4600 yr BP were recounted to differentiate at least 125 well-preserved *Pinus* pollen grains per sample, from which ratios of the pollen morphotypes (*P. Subg. Strobilus*; *Pinus Subg. Pinus*) were calculated (Appendix D).

4. Results

4.1. Pollen and macrofossil sequences at Tower Lake

Here we present data for the eight primary arboreal pollen taxa at Tower Lake (*Picea*, *Abies*, *Pinus*, *Betula*, *Quercus*, *A. saccharum*,

Tsuga, *Fagus*) and the eleven primary arboreal taxa represented in the corresponding macrofossil record, using the most abundant organ-type for each (*Picea* needles, *A. balsamea* needles, *P. banksiana* needles, *P. resinosa* needles, *P. strobus* bud scales, *B. papyrifera* fruits and bracts, *Betula alleghaniensis* fruits and bracts, *Q. rubra* bud scales, *A. saccharum* bud scales, *T. canadensis* needles, *F. grandifolia* bud scales) (Fig. 3). The full data sets will be archived in the Neotoma database (<http://www.neotomadb.org/>).

4.1.1. Zone T-1a (8500–7500 yr BP)

Pollen assemblages (8500–7500 yr BP) are dominated by *Pinus* (50–75%) and *Betula* (15–35%), with substantial *Picea* (2–4%) and *Abies* (1–3%) (Fig. 3). *Pinus* pollen includes both *Pinus* Subgenus *Pinus* and *P. strobus*. *Picea* needles occur in sediments ≥8300 yr BP, together with macrofossils of *Abies*, *P. banksiana*, *P. resinosa*, and *B. papyrifera*. *P. strobus* bud scales occur after ca 8300 yr BP.

4.1.2. Zone T-1b (7500–6500 yr BP)

Picea and *Abies* pollen decline to trace percentages 7500 yr BP; *Abies* needles occur sporadically until 6100 yr BP (Fig. 3). *P. resinosa* needles occur frequently, although *Pinus* Subgenus *Pinus* pollen is low (<3%). *P. strobus* bud scales are abundant from ca 8000 to 6000 yr BP, and *P. strobus* pollen increases between 7000 and 6500 yr BP. *Betula* pollen decreases to <20% after 7200 yr BP, and *Quercus* pollen increases to ≥5% after 7000 yr BP. *B. papyrifera* macrofossils occur consistently throughout this interval, and *Q. rubra* bud scales are present in many macrofossil samples after 7000 yr BP.

4.1.3. Zone T-2 (6500–5000 yr BP)

Tsuga pollen first appears in the record ca 6300 yr BP, and increases rapidly starting ca 6000 yr BP to a maximum (>20%) by 5700 yr BP (Fig. 3). *T. canadensis* needles first occur 6300 yr BP, and are both frequent and abundant by 6000 yr BP. *A. saccharum* pollen increases beyond trace amounts ca 6300 yr BP, coincident with first occurrence of *A. saccharum* bud scales.

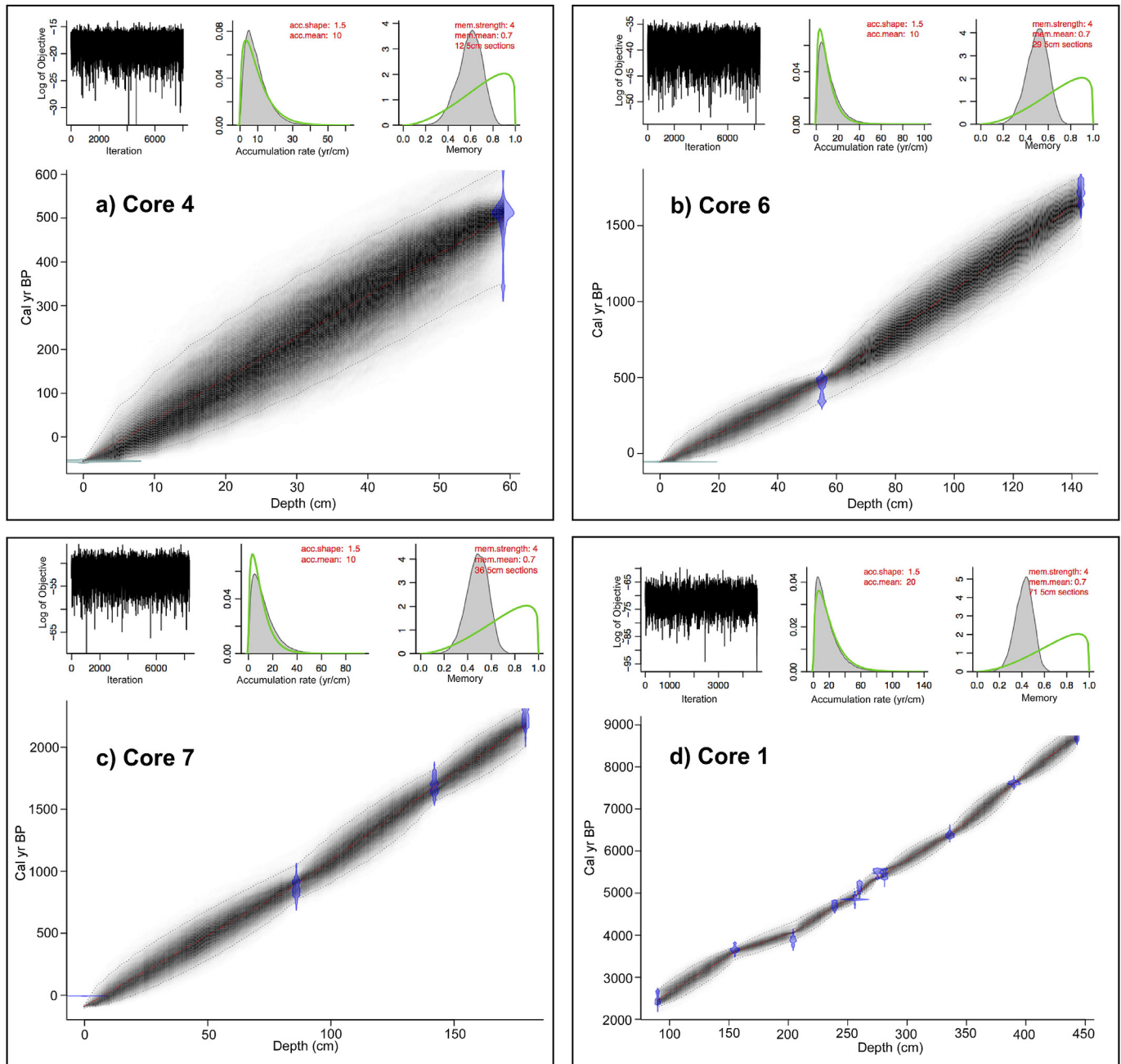


Fig. 2. Bacon age–depth models (Blaauw and Christen, 2011) for Tower Lake cores: (a) Core 4, (b) Core 6, (c) Core 7, (d) Core 1. For each age-model, insets in the upper panel show the iteration history and two parameters of the Bayesian age-model. The left inset shows the iteration history, the middle inset shows the prior (green line) and posterior (gray area) of the sediment accumulation rate (yr/cm), and the right inset shows the prior (green line) and posterior (gray area) of the memory (1 cm autocorrelation strength). The bottom panels show the age-depth models with uncertainties. Solid red lines denote weighted averages of all possible chronologies. Associated uncertainties are represented by the gray-scale and confidence intervals (dotted black lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

A transient *Tsuga* pollen minimum (<9%) is dated ca 5500 yr BP. *Tsuga* needles are abundant in most samples dating between 6100 and 4900 yr BP, although two minima occur. The first, brief minimum coincides with the pollen minimum ca 5500 yr BP. The second, more extended needle minimum coincides with a decrease in *Tsuga* pollen between 5200 and 5100 yr BP (Fig. 3). *Pinus* pollen decreases from 60 to 70% to less than 50% between 6000 and 5500 yr BP, and remains relatively low (40–50%) until after 5000 yr BP. *P. strobus* macrofossils are extremely abundant between 7500 and 5900 yr BP, but decrease dramatically (ca 1/5 of previous

abundance) after 5900 yr BP, coinciding with the maximum of *Tsuga* macrofossils and with the *Pinus* pollen minimum. Needles of *P. resinosa* occur sporadically until 5600 yr BP, when they disappear from the record. Pollen ratios of *P. Subg. Pinus* to *P. Subg. Strobus* vary little (Appendix D).

Betula pollen is low, rarely exceeding 15%, between 6500 and 5000 yr BP. It is particularly low ($\leq 5\text{--}6\%$) from ca 6000 to 5600 yr BP. *B. papyrifera* macrofossils occur in low numbers between 6500 and 5000 yr BP, and are absent from many samples dating between 6000 and 5500 yr BP. *A. saccharum* and *Quercus* pollen show

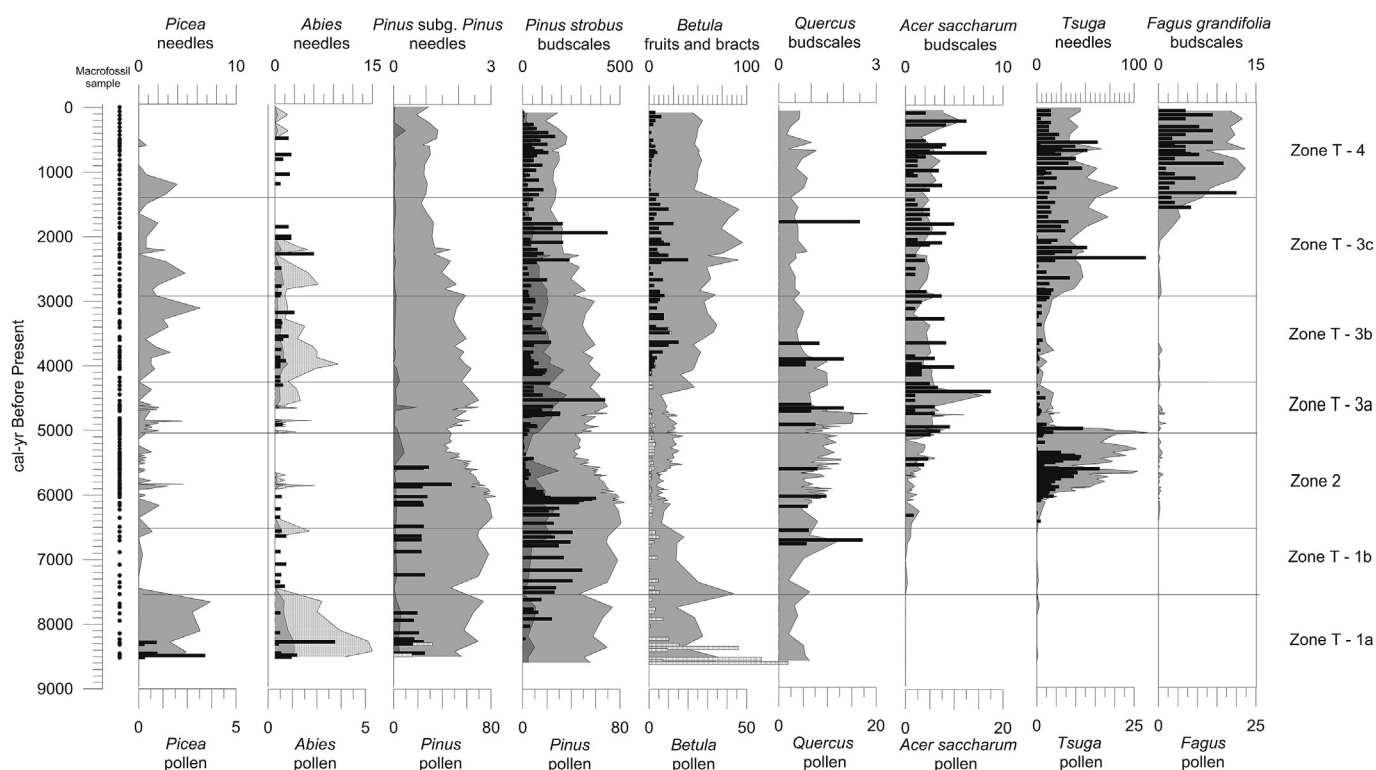


Fig. 3. Pollen (silhouettes) and macrofossil abundances (horizontal bars) for selected tree taxa and macrofossil types at Tower Lake. Pollen data are expressed as percentages, and macrofossil data represent numbers per 50 cm³ sediment volume. Note that Total *Pinus* pollen is portrayed twice; dark silhouettes for *Pinus* pollen denote *Pinus* Subg. *Pinus* in left-hand profile and *P. strobus* in right-hand profile. Black bars in *Pinus* Subg. *Pinus* macrofossil profile denote *P. resinosa*; dotted bars denote macrofossils of *P. banksiana*. Black bars in *Betula* macrofossil profile indicate *Betula alleghaniensis* macrofossils; dotted bars portray *B. papyrifera*. Light silhouette for *Abies* pollen represents 5× exaggeration.

fluctuating but increasing abundance after 6000 yr BP, and *A. saccharum* and *Q. rubra* macrofossils occur intermittently throughout the period.

4.1.4. Zone T-3a (5000–4200 yr BP)

Tsuga pollen declines sharply 5050 yr BP, and maintains relatively constant low levels (3–5%) until 4200 yr BP (Fig. 3). *Tsuga* needles are frequently absent and never abundant during this period (Fig. 3). The *Tsuga* decline is initially followed by a transient maximum in *Quercus* pollen, a more persistent maximum of *Pinus* pollen, and a maximum of *A. saccharum* pollen (which begins before the *Tsuga* decline). *P. strobus* bud scales increase in number almost immediately after the *Tsuga* decline. *B. papyrifera* macrofossils show no increase after the *Tsuga* decline, and are absent from many samples dating between 4700 and 4200 yr BP. *Betula* pollen percentages are also low 4800–4200 yr BP.

4.1.5. Zone T-3b (4200–2800 yr BP)

Betula pollen increases sharply at 4200 yr BP. *B. papyrifera* macrofossils occur consistently in samples dating between 4150 and 3900 yr BP (Fig. 3), and are absent from nearly all younger samples. *B. alleghaniensis* macrofossils occur in one sample dating to 4600 yr BP, but they occur in all samples after 4050 yr BP, usually in higher abundance than attained by *B. papyrifera* macrofossils in the preceding 4000 years.

After the rapid increase 4200 yr BP, *Betula* pollen increases more or less steadily to ca 30% by 3500 yr BP, and *Pinus* pollen decreases. *Quercus* pollen has a brief maximum between 4200 and 3800 yr BP. *Q. rubra* bud scales are rare after 3600 yr BP. *Abies* macrofossils are frequent from 4200 to 3200 yr BP. *A. saccharum* pollen is relatively constant between 4200 and 1700 yr BP, and bud scales occur

frequently throughout the period. *Tsuga* pollen undergoes a transient decline to <3% between 4200 and 3000 yr BP, and needles continue to be infrequent.

4.1.6. Zone T-3c (2800–1400 yr BP)

Tsuga pollen increases ca 2800 yr BP, reaching a maximum of 10–15% by 2600 yr BP (Fig. 3). *Tsuga* needles also increase in frequency and number; they occur in every macrofossil sample ≤3300 yr BP and are particularly abundant after 2800 yr BP (Fig. 3). *Pinus* pollen continues to decline from 2800 to 1400 yr BP, although *P. strobus* bud scales continue to be abundant. *Betula* pollen attains a maximum (>35%) during this period, and *B. alleghaniensis* macrofossils are abundant in all samples. *A. saccharum* pollen remains well-represented (3–5%), and bud scales occur in about half of macrofossil samples in the interval.

4.1.7. Zone T-4 (1400–55 yr BP)

Betula pollen declines sharply to <30% 1400 yr BP, and *B. alleghaniensis* macrofossils drop in abundance at about the same time (Fig. 3). *Fagus* pollen, which occurs intermittently in trace amounts after 2800 yr BP, increases steadily after 1800 yr BP and reaches a maximum ≥15% after 1000 yr BP. *Fagus* bud scales first occur 1500 yr BP and are present in nearly all younger samples. *Tsuga* pollen declines modestly to ≤15% 1000 yr BP, though macrofossils remain abundant. *Pinus* pollen, *P. strobus* bud scales, and *A. saccharum* pollen and bud scales show little change.

4.2. Comparison of macrofossil morphotypes within taxa

Correspondence between the macrofossil sequences and pollen sequences at Tower Lake is most obvious at coarser (centennial and

above) temporal scales, where macrofossil presence/absence, frequency, and abundance generally track abundance patterns in the pollen data (Fig. 3). In the mid-Holocene section of contiguous pollen and macrofossil sampling (6100–4600 yr BP), macrofossils and pollen have parallel patterns at finer, subcentennial scales. Striking examples include (1) the *P. strobus* bud-scale minimum 6000–5000 yr BP, during a *Pinus* pollen minimum, (2) the two *Tsuga* macrofossil minima that coincide with transient pollen minima at 5500 and 5200 yr BP, (3) the terminal *Tsuga* decline at 5000 yr BP, and (4) the paucity of *Betula* macrofossils during *Betula* pollen minima 6000–5500 and 4700–4200 yr BP.

In order to evaluate whether these correspondences are coincidental, we compare these observed macrofossil patterns with those of other organ types for the same tree taxa. Here we concentrate on the following comparisons between primary types used in Fig. 3 (bold) and other types: *Picea*: **needles** vs. bud scales; *A. balsamea*: **needles** vs. cone-scale bracts; *P. resinosa*: **needles** vs. bud scales, short shoots and microstrobili; *P. strobus*: **bud scales** vs. needles, short shoots, microstrobili, and seeds; *Betula papyrifera*: **fruits** vs. **catkin bracts**; *B. alleghaniensis*: **fruits** vs. **catkin bracts**; *A. saccharum*: **bud scales** versus samaras and seeds; *T. canadensis*: **needles** vs. twigs, microstrobili, and seeds. *Q. rubra* and *F. grandifolia* were represented only by bud scales, so were not included in this comparison. Both *Betula* species were represented only by fruits and catkin bracts. We chose to plot them together in Fig. 3 to draw out temporal patterns, but here plot them separately for comparison, together with undifferentiated specimens to assess whether preservation quality affected our results.

Picea needles and bud scales show similar patterns of high abundance before 8500 yr BP, followed by absence from the rest of the record (Fig. 4). *A. balsamea* needles are also most abundant before 8500 yr BP, as are cone-scale bracts (Fig. 4). However, bracts are absent from all younger sediments, whereas *Abies* needles occur in low numbers but high frequency through the rest of the Holocene, except the period of *Tsuga* dominance 6000–5000 yr BP. The absence of bracts younger than 8500 yr BP may indicate greatly reduced numbers of adult *Abies* trees near the lake margin; adult trees produce many more needles than cone-scales, and juvenile individuals also produce needles. Patterns of *P. resinosa* needles contrast with those of other organ types (Fig. 4). Needles occur frequently but in low numbers between 8500 and 5600 yr BP, whereas bud scales, short shoots and microstrobili are abundant only before 8000 yr BP. As in the case of *Abies*, this may indicate reduction of tree populations, with consequent reduction of macrofossil flux density to the lake.

Patterns among organ types for *P. strobus* are generally consistent (Fig. 5). All show absence before 8300 yr BP, high frequency and abundance 7500–6000 yr BP and after 5000 yr BP. All show low frequency and abundance during the *Pinus* pollen minimum 6000–5000 yr BP, and a decline in abundance 3500–3000 yr BP (Fig. 5). The increase in bud scales after 2500 yr BP coincides with the transition from Core 1 to Cores 6 and 7; macrofossil concentrations of other types (*Tsuga* needles, *B. alleghaniensis* fruits and bracts) all increase at the same point. In some individual samples, the concentration of one *P. strobus* morphotype is anomalously high compared to the same morphotype in adjacent samples, and to other morphotypes in the same sample. Examples include the unusually large bud-scale spikes ca 1860 and 4440 yr BP, the microstrobilus spike ca 1790 yr BP, the short-shoot peak ca 4440 yr BP, and the unusually high needle concentrations in two samples 5200–5300 yr BP (Fig. 5). These presumably represent singular taphonomic events (e.g., individual or limb mortality of an overhanging tree).

All *Tsuga* macrofossil types are rare or absent before 6100 yr BP, frequent and abundant from 6000 to 5500 yr BP, scarce between

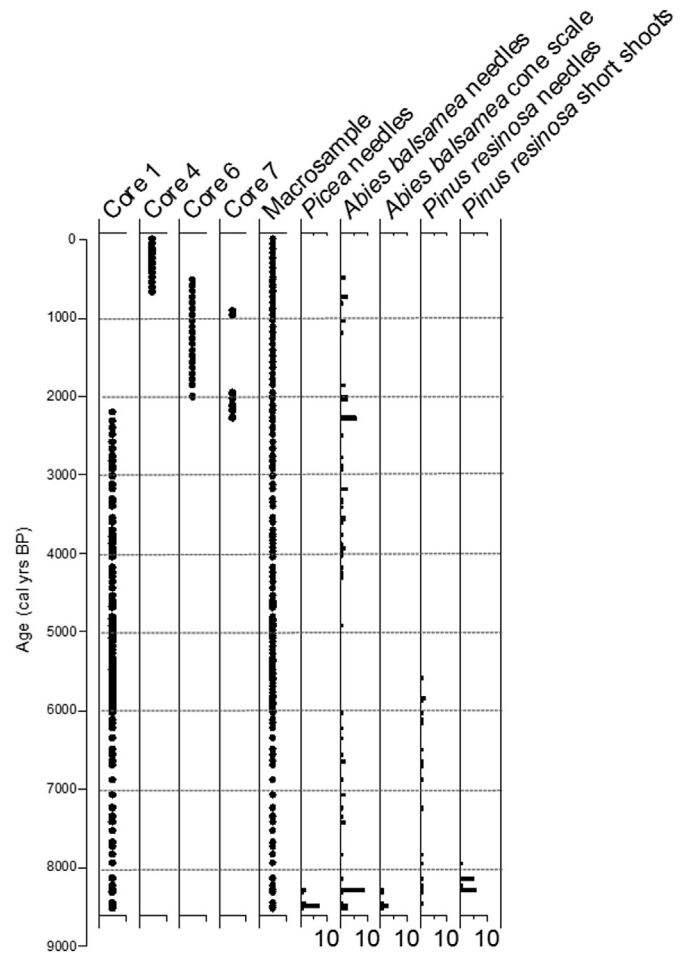


Fig. 4. Temporal sequence of abundance of macrofossil types for *Picea* sp., *Abies balsamea*, and *Pinus resinosa* from Tower Lake. Abundances are expressed as number of macrofossils per 25 cm³.

4800 and 3000 yr BP, and frequent and abundant after 3000 yr BP (Fig. 6). Although macrofossil types other than needles are not present in all samples, they show generally similar fine-scale patterns. In particular, the transient needle minima centering on 5400 and 5200 yr BP are accompanied by minima in bud scales, short shoots, microstrobili, and seeds (Fig. 6). All organ-types declined abruptly at the time of the *Tsuga* pollen decline, and the *Tsuga* pollen minimum 4200 yr BP is associated with absence or scarcity of all *Tsuga* macrofossil types. Brief declines in all *Tsuga* organ types also occur 2400 and 1950 yr BP, suggesting transient population declines; the decline at 1950 yr BP is accompanied by a brief pollen minimum (Fig. 3). Macrofossils of other taxa are well-represented in these intervals (Figs. 3 and 5).

A. saccharum samaras and seeds are less frequent in sediments than bud scales, but all occurrences are within the period of bud-scale representation (Fig. 7). Samaras and seeds are most frequent and abundant between 4700 and 4200 yr BP.

Fruits and bracts of *Betula* are not strictly independent; each bract bears two fruits, and all are dispersed aerally once the catkin dehiscence and disintegrates. The seeds are much smaller, lighter, and are winged, so dispersal properties of the two types should differ. Patterns for the two organ types are similar for both species. *B. papyrifera* fruits and bracts are abundant before 8000 yr BP, frequent but in low numbers from 8000 to 4500 yr BP, and absent or rare after 4200 yr BP (Fig. 7). *B. alleghaniensis* fruits and a bract

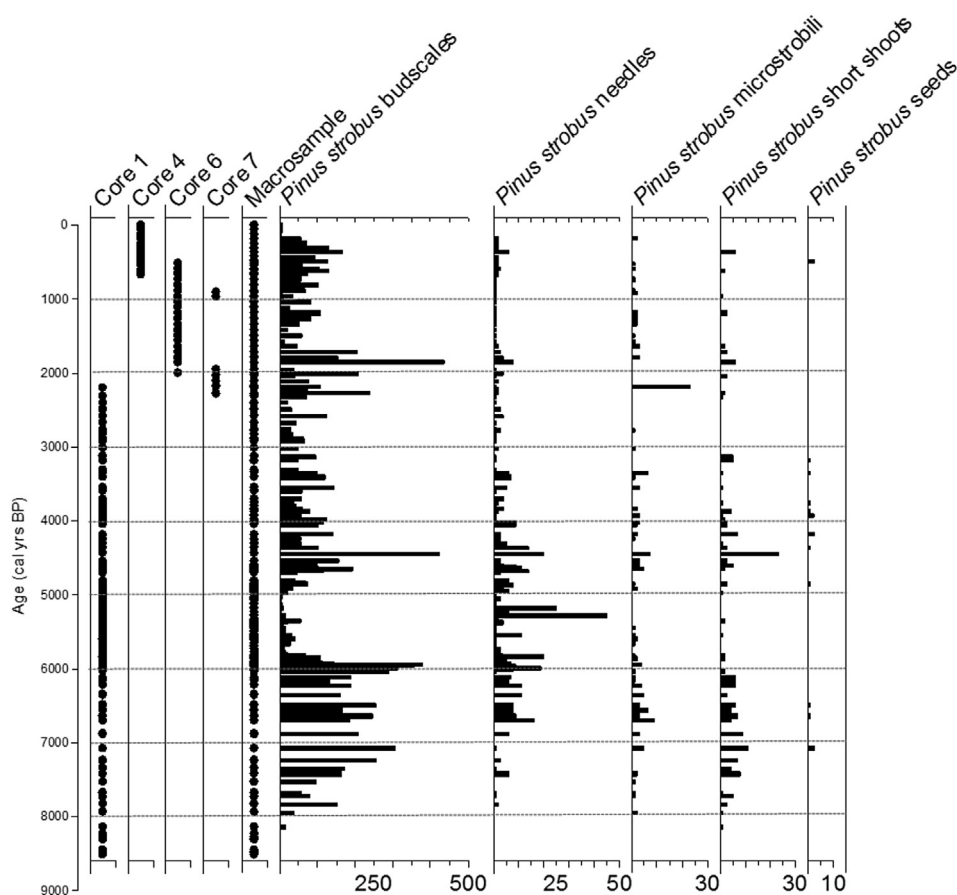


Fig. 5. Temporal sequence of abundance of macrofossil types for *Pinus strobus* from Tower Lake. Abundances are expressed as number of macrofossils per 25 cm³.

occur in a single sample dating to 4600 yr BP, and both are frequent and abundant after 4100 yr BP. Both organ-types show an increasing pattern between 4100 and 3800 yr BP. Unidentifiable bracts (usually decayed and eroded) occur sporadically through the sediment sequence. In contrast, unidentifiable *Betula* fruits (seeds and styles intact, but wings completely missing) are almost entirely restricted to the early Holocene record, and are most abundant during the *B. papyrifera* peak more than 8000 yr BP (Fig. 7). Thus, the apparent decline in *B. papyrifera* after 8000 yr BP, and the minima during the period of *Tsuga* dominance and after the *Tsuga* decline, are not artifacts of preservation or identifiability.

5. Discussion

5.1. Using macrofossils to infer changes in local forest composition

Temporal changes in macrofossil abundance might represent changes in local population size of trees, statistical variation owing to low sample size, or variations in taphonomic processes (dispersal, deposition, preservation). Deposition- and preservation-related taphonomic influences are minimized when sediment lithology, deposition rates, and total macrofossil concentrations are relatively constant, as is often the case for Holocene lake sediments. When different organ-types within a taxon show congruent temporal patterns of abundance, those patterns should indicate changes in local population size of the taxon or changes in taphonomic processes that affect multiple organ-types. The case for inferring changes in population size is strengthened if the various organ-types for an individual taxon differ substantially in size,

mass, buoyancy, or tissue composition. Such differences should impose different taphonomic biases, including dispersal to the lake and dispersal and deposition within the lake. Therefore, temporal congruence of different macrofossil types should indicate that changes in population size have overridden variations in dispersal and deposition. Furthermore, if the different organ-types represent different functions (e.g., photosynthesis versus sexual reproduction), temporal coherence should dampen two more influences: (a) changes in age-structure of the population (e.g., seedlings and saplings produce foliage and bud-scales without producing seeds or microstrobili), and (b) changes in biomass allocation within individuals (e.g., dampening or amplification of seed production relative to photosynthetic and structural tissues).

Our data show temporal correspondence between different organ-types within all the tree species examined (Figs. 4–7). Patterns are particularly strong for *P. strobus* and *T. canadensis*, where each species is represented by multiple organs supporting reproduction (seeds, microstrobili) and photosynthesis (needles, bud scales, twigs, short shoots) (Figs. 5 and 6). Fewer macrofossil types are represented for *A. balsamea*, *P. resinosa*, and *A. saccharum*, but each species includes both reproductive and foliar/support organs that show parallel temporal patterns, albeit at coarser temporal scales (Figs. 4 and 7). For example, *A. saccharum* bud scales and seeds occur in low concentrations and are absent from many samples, but nearly all occurrences are between 5500 yr BP and the present, and both organ types are infrequent between 3800 and 2000 yr BP (Fig. 7). *Betula* spp. are represented only by fruits and ovulate catkin-bracts, which are functionally related (each bract bears two fruits), and so our comparison is weaker than for other

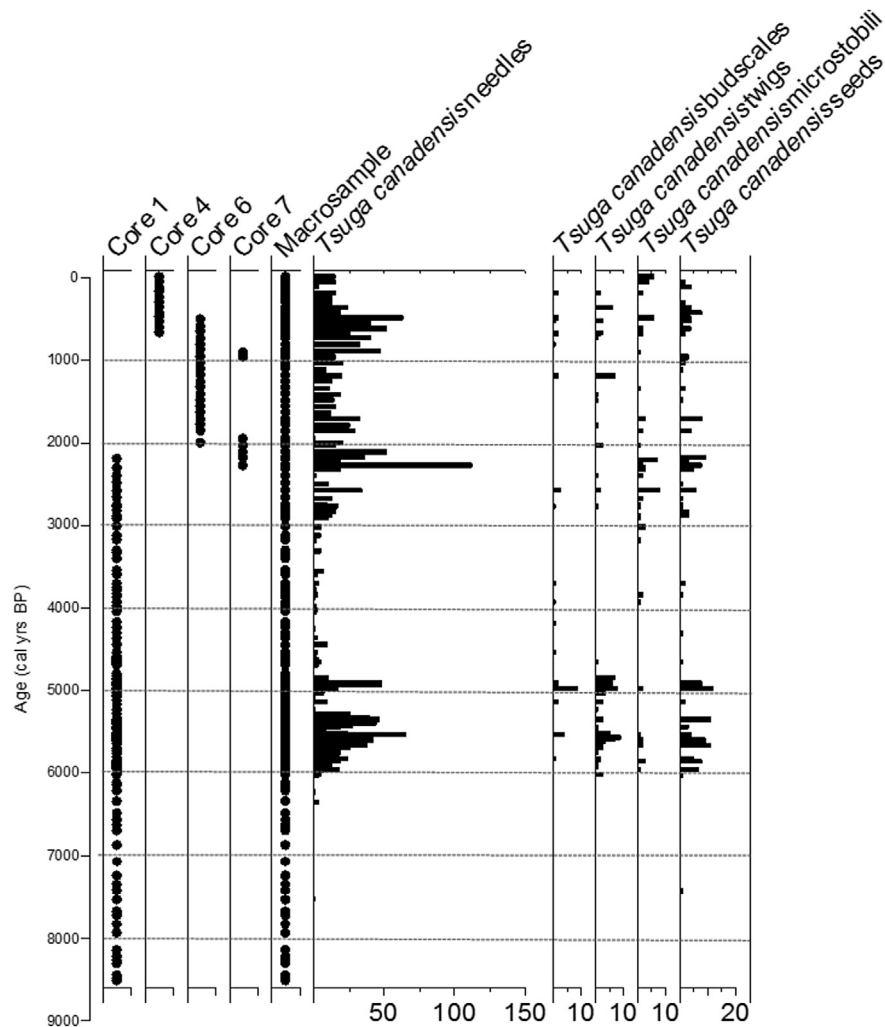


Fig. 6. Temporal sequence of abundance of macrofossil types for *Tsuga canadensis* from Tower Lake. Abundances are expressed as number of macrofossils per 25 cm³.

species. Nonetheless, fruits and bracts differ in dispersal, deposition, and preservation properties, and their temporal correspondence supports the inference that macrofossil concentrations track reproduction and, indirectly, abundance of nearby *Betula* trees (Fig. 7).

The temporal coherence among organ types within taxa indicates that macrofossil concentrations in Tower Lake sediments have recorded changes in size and proximity of populations of individual tree species. Similar coherence is visible in other macrofossil records (e.g., Jackson, 1989; Spear, 1989; Lotter, 1999; Wick et al., 2003; Finsinger and Tinner, 2007). Lake-sediment macrofossil records, from carefully selected sites and with high sample sizes, can support broad inferences concerning local presence and abundance of species in adjacent forests, and changes in abundance relative to other time-periods. Estimating the absolute magnitude of change in tree abundance is more difficult. For example, a five-fold increase in *Tsuga* needles per unit volume might represent a roughly equivalent increase in *Tsuga* tree abundance (density, basal area, foliar density) but might also be influenced by greater proximity of individuals to the lakeshore.

Application of statistical calibrations of macrofossil abundance in lakes with nearby trees may be elusive, because of the sensitivity of macrofossil concentrations to lake size, morphometry, and other taphonomic factors (Jackson and Booth, 2007). It seems unlikely

that a single calibration model could be applied to multiple sites except under extremely constrained circumstances (e.g., virtually identical morphometry). In some cases, macrofossil percentages can allow comparison of macrofossil abundances among sites (Watts and Winter, 1966; Dunwiddie, 1986; Jackson and Whitehead, 1991; Tinner and Theurillat, 2003). In a unique calibration study of six coniferous tree species, Dunwiddie (1987) observed statistically significant positive relationships between percent macrofossil abundance and percent tree abundance within 30 m of 30 small ponds and lakes (≤ 0.1 –44.7 ha; median 1.4 ha) in Washington. Statistical significance for many species, however, was leveraged by sites with neither trees nor macrofossils, and all relationships showed considerable scatter. Although the general relationships may hold for ponds of similar size, application to larger lakes may be problematic because of differences among taxa in dispersibility and buoyancy. Blarquez et al. (2012) developed an empirical calibration between needle accumulation rates and basal area estimates for two coniferous trees using litter traps in forest openings, and applied them to Holocene macrofossil data from two small lakes. This approach provided a formal means to adjust abundance estimates of the two taxa (one much more productive of needles than the other). Complications may arise from differences in airborne dispersal, flotation, imbibition and sinking rates, and other taphonomic factors. Although macrofossil-vegetation

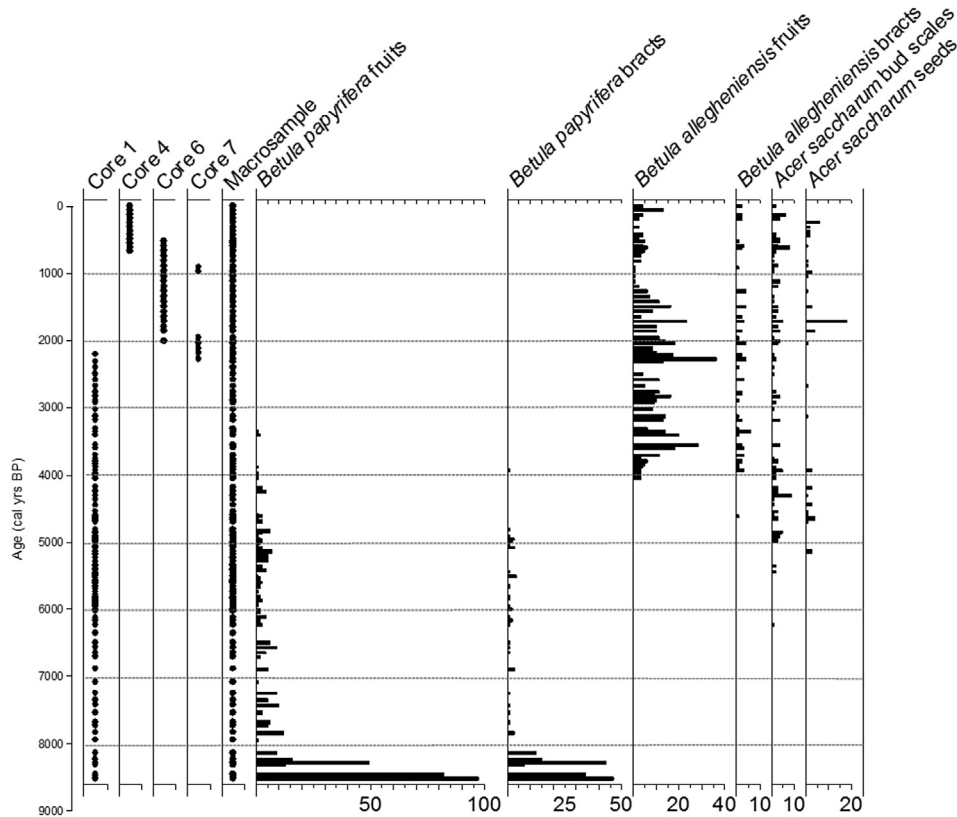


Fig. 7. Temporal sequence of abundance of macrofossil types for *Betula papyrifera*, *B. alleghaniensis*, and *Acer saccharum* from Tower Lake. Abundances are expressed as number of macrofossils per 25 cm³.

calibrations may have limited application, comparative studies (Birks, 1973; Drake and Burrows, 1980; Dunwiddie, 1987; Blarquez et al., 2012) are useful in developing and validating informal rules and judgments for drawing inferences (Jackson, 2012).

Interpretation of both pollen and macrofossil profiles often proceeds in an informal, judgmental fashion, in which the interpreter (author, reader, downloader) evaluates the significance of temporal patterns, applying informal weightings based on judgment and experience (prior knowledge). Taken into account are the magnitudes, rates, and durations of changes, sample size, and various taphonomic and representation factors. The interpreter chooses to ignore some changes as insignificant, while identifying others as signifying actual changes in abundance in vegetation. Inferences are reinforced when they 'make sense' in broader context – in the case of pollen, when the underlying patterns are observed at other sites, when other records (e.g., macrofossils) provide corroboration, and when inferences are consistent with prior ecological and climatological knowledge.

Interpretation of plant-macrofossil sequences may never be as routine as pollen interpretation; macrofossil abundances are much more variable within and among lakes. Pollen percentages provide a single common currency that allows comparison among assemblages, sites, and depositional environments. In contrast, macrofossils are expressed in multiple, idiosyncratic currencies that are often site-specific, and that can even change within a site as depositional or other taphonomic conditions change. Nonetheless, the information retained in macrofossil sequences can be used more effectively by developing a better empirical and conceptual understanding of macrofossil assemblages and the factors underlying them. Such understanding can be made explicit by building formal conceptual models of macrofossil representation (Jackson,

2012), and enriched by incorporating natural history, field observations, targeted studies of taphonomic processes, and comparative studies of modern macrofossils and vegetation.

5.2. Linking local and regional vegetational change using macrofossil and pollen data

Pollen and macrofossil assemblages are under different taphonomic influences, and are biased in different, often-contrasting ways. There is no *a priori* reason for pollen and macrofossil sequences to align with each other unless they are recording the same fundamental phenomena, albeit at different spatial scales. Close correspondence between pollen and macrofossil sequences for specific taxa, as observed in the Tower Lake record, can arise only under specific circumstances. In particular, the local vegetation recorded by macrofossils must be representative of the regional vegetation that dominates the pollen record; in essence, the vegetation near the lake-margin must undergo similar changes, under similar controls, at similar times, as the aggregated vegetation over the surrounding region. This is likely only if the local environment adjacent to the lake is representative of the surrounding region. Under such circumstances, the macrofossil record can be viewed as a plot- or stand-scale sample, one of hundreds to thousands of similar individual stands that aggregate, together with other stands of different composition, to yield the pollen record in the lake sediments. The macrofossil record then serves as a means for understanding some of the local ecological dynamics underlying the pollen record, albeit based on a single quasi-plot.

The local vegetation sampled by plant macrofossils is a biased representation of the regional vegetation, simply because the environment near a lakeshore has unique edaphic and topographic

properties relative to surrounding uplands. In Fig. 1, for example, conifers (primarily *T. canadensis* and *P. strobus*) are concentrated along the edge of Tower Lake, and are more sparse in uplands more than 20–50 m from the lake. The contrast between near-shore and upland composition may have been amplified in the past century as a legacy of logging and other disturbances; the region has undergone a net decline in conifers since the early 19th Century (Schulte et al., 2007). Upland old-growth remnants in Upper Michigan west of Tower Lake have *T. canadensis* as a co-dominant, with local occurrence of *P. strobus* (Frelich et al., 1993; Woods, 2000a, 2000b), and presettlement vegetation in central Upper Michigan and adjacent Wisconsin had substantial representation of *Tsuga* and *Pinus* (He et al., 2000; Albert and Comer, 2008). Nevertheless, a number of environmental features (e.g., cold-air drainage, higher incident sunlight) may have biased forest composition on the slopes adjacent to our coring site relative to unbroken uplands during the Holocene. Regardless of any quantitative bias in forest composition between the two proxies, close temporal correspondence of the kind seen in the Tower Lake record indicates that local changes in abundance correspond to regional changes.

Other studies, particularly studies of small lakes in conifer-rich forest regions, show parallel temporal patterns in pollen and macrofossil abundances (Anderson et al., 1986; Dunwiddie, 1986; Wainman and Mathewes, 1987; Jackson, 1989; Spear, 1989; Yu, 1997; Lotter, 1999; Barnekow and Sandgren, 2001; Jackson and Booth, 2002; Wick et al., 2003; Finsinger and Tinner, 2007; Gavin et al., 2013). The degree of correspondence varies among sites and among taxa at individual sites. Discrepancies arise for a number of reasons, including mismatch between local and regional vegetation composition. In some instances, macrofossil increases lag behind pollen increases for particular taxa, owing to pollen transport from advancing but still-distant populations (e.g., Jackson, 1989). Pollen at high-elevation sites in mountainous regions is often biased toward lower-elevation vegetation (Jackson, 1989, 1991; Spear, 1989; Jackson and Whitehead, 1991; Fall, 1992; Jackson and Smith, 1994; Pellatt and Mathewes, 1994, 1997; Spear et al., 1994), which usually occupies a much greater part of the surrounding landscape. Also, because occupation of all suitable local habitats by an invading woody species may require decades to centuries (e.g., Gray et al., 2006; Lesser and Jackson, 2012), lags may be expected between regional expansion and local appearance. This may account, for example, for the apparent lag between the *Fagus* pollen increase and the first occurrence of macrofossils at Tower Lake (Fig. 2).

Even when local and regional vegetation are likely to be similar, correspondence may be imperfect or obscure owing to sampling issues (low macrofossil sample size; low temporal density of macrofossil or pollen sampling; poor representation of certain pollen types) and depositional and other taphonomic factors. Individual samples may show unusually high abundances of particular macrofossils with no corresponding pollen peak. Many of these undoubtedly derive from local, often one-off events (e.g., needle shedding from mortality of a tree overhanging the lake; dropping of a needle- or seed-laden twig, branch, limb, or entire tree into the lake). Such anomalies at Tower Lake might include the *Tsuga* needle maximum 2275 yr BP, *P. strobus* budscale maxima 1855 and 4440 yr BP, and the *P. strobus* needle events 5200–5300 yr BP (Figs. 3 and 5).

Tandem application of pollen and plant macrofossil sequences to differentiate local and regional vegetation has considerable potential, largely unrealized, for linking local ecological dynamics to regional patterns. Multiple sites representing similar local habitats can be used to assess spatio-temporal patterns and synchrony of local events, which can help understand the site-specific ecological dynamics underlying changes in the pollen record. Macrofossil sequences from sites in contrasting edaphic situations can assess

effects of climate change and regional species pools on vegetational heterogeneity across the landscape, adding spatial detail to build on coarser pollen studies (e.g., Brubaker, 1975; Ford, 1990; Ewing, 2002). In a recent study, Gavin et al. (2013) applied pollen and macrofossils together to examine local and regional disturbance dynamics; this approach can be extended to multiple sites within a region. Finally, macrofossil sequences can validate and complement applications of formal approaches to spatially explicit vegetational inferences (e.g., Sugita et al., 2010). Multi-site studies, comprising pollen records from lakes of different sizes (Sugita, 2007a, 2007b) and paired pollen-macrofossil records from selected small lakes could provide a powerful means for revealing vegetation patterns and dynamics across a range of spatial scales.

5.3. Holocene forest dynamics: multiple processes, multiple scales

To a first approximation, the Tower Lake represents forest community response to a series of climatic changes, some gradual and others rapid, with new species becoming established as local climate became suitable. The major features of the record, including the early *Pinus* maximum, the replacement of *Picea* and *P. banksiana* by *P. strobus*, the rise of *Quercus* and *A. saccharum*, the *Tsuga* maximum (6000–5000 yr BP), minimum (5000–3000 yr BP), and resurgence (3000–0 yr BP), and the late Holocene *Fagus* increase, are generally attributable to climatic changes, which are indicated in pollen, lake-level, and peatland records from the region (Bartlein et al., 1984; Singer et al., 1996; Davis et al., 2000; Booth et al., 2004; Shuman et al., 2004; Williams et al., 2010; Shuman, 2012). The local and regional dynamics respectively revealed in the macrofossil and pollen records at Tower Lake scale up to the broad-scale biogeographic and vegetational responses to climate change visible at subcontinental scales (Webb, 1986, 1987; Jackson et al., 1997; Williams et al., 2004).

These broad-scale syntheses and climatic attributions, however useful, are necessarily incomplete, because they overlook both the proximal ecological mechanisms (mortality, disturbance, recruitment, population growth, resource competition, consumption) that underlie these changes (Jackson and Blois, in revision) and the finer-scale decadal to submillennial climate changes and variability that shape community structure by modulating and redirecting successional mechanisms and pathways (Jackson et al., 2009). The spatially, temporally, and taxonomically precise record from Tower Lake (Fig. 3) provides some glimpses of those finer-scale dynamics, even if those dynamics remain largely unexplained in detail. For example, Pollen Zone T-2 (6500–5000 yr BP), which includes the *Tsuga* rise and maximum, comprises an overlapping series of vegetational transitions. Starting with the rapid increase in *Tsuga* pollen and macrofossils 6000 yr BP, *Pinus* pollen commenced a slow, steady decline that continued until the *Tsuga* decline 5000 yr BP (Fig. 3). *P. strobus* macrofossils decreased rapidly 5900 yr BP, becoming scarce until 5000 yr BP. *Betula* pollen was very low, and *B. papyrifera* macrofossils scarce, 6000–5500 yr BP, and *A. saccharum* pollen was low (and macrofossils absent) during this same period. Pollen of *Betula*, *Quercus*, and *A. saccharum* all increased after the first, transient decline of *Tsuga* pollen 5500 yr BP.

Some of the subcentennial to submillennial dynamics in Pollen Zone T-2 and other parts of the profile may represent responses to subtle climatic changes or transient climatic events. For example, the *Tsuga* ‘minimum-within-a-minimum’ 4200–3900 yr BP coincides with a severe, persistent drought event recorded at South Rhody Bog, 4 km northwest of Tower Lake (Booth et al., 2005). Forest dynamics at multidecadal to submillennial scales, however, probably do not represent simple cause-effect responses to climate change. Population and community processes play particularly prominent roles at these finer timescales; climate events and

transitions can reset or redirect such ecological processes as mortality, recruitment, interspecific competition, and invasion (Swetnam and Betancourt, 1998; Jackson et al., 2009; Jackson and Blois, in revision; Pedersen et al., 2014). Herbivores, pathogens, and wildfires can have similar effects, in some cases interacting with climate to reset successional clocks and alter successional trajectories. In forests, these processes may take decades or centuries to run their course, during which they may be redirected by additional climatic events or transitions.

Paleoecologists working with records exceeding more than a few decades or centuries are confronted with a gargantuan explanatory task, because underlying their records is a multitude of mechanisms operating and interacting across a wide range of spatial and temporal scales. The challenge of understanding cross-scale interactions is not unique to paleoecology (e.g., Peters et al., 2004, 2007), but paleoecologists confront a broader range of relevant temporal (and sometimes spatial) scales than most ecologists. Developing conceptual frameworks, sampling designs, and analytical frameworks capable of addressing cross-scale interactions is an important challenge for paleoecologists, because it addresses the mechanisms by which climatic and other environmental changes lead to ecological and biogeographic changes. Paleoecology is unique in its ability to provide long-term observations of such dynamics in a variety of settings (NRC, 2005; Jackson and Blois, in revision).

Application of ecological models to paleoecological problems represents one potentially fruitful approach to meeting this challenge. For example, dynamic global vegetation models (DGVMs) and forest simulation models can be driven by transient climate-change models to explore mechanisms and test hypotheses. Such models can incorporate higher-frequency (interannual to centennial) climate variability, as well as high-frequency ecological mechanisms. Recent applications are promising (Hickler et al., 2004; Miller et al., 2008), but challenges remain, particularly translating between paleoecological data and ecological properties.

A complementary approach is to focus attention on specific vegetational or climatic events or transitions, particularly with multiple proxies (e.g., Gray et al., 2006; Hotchkiss et al., 2007; Gill et al., 2009, 2012; Tveit et al., 2009; Shuman et al., 2009b; Booth et al., 2012a, 2012b). Such studies integrating highly resolved records with independent proxies for climate and other influences allow close examination of ecological mechanisms in context of environmental change, revealing complex cross-scale interactions. They can also leverage increased capacity for high-precision dating and statistical assessment of synchrony within and among sites (e.g., Booth et al., 2012a). Individual high-resolution records such as Tower Lake can identify 'critical intervals' from which much can be learned about cross-scale interactions, and any of several might be selected. Because of the widespread attention given to the *Tsuga* decline (Fuller, 1998; Foster et al., 2006; Shuman et al., 2009a; Oswald and Foster, 2011; Liu et al., 2012; Booth et al., 2012a; Marsicek et al., 2013), the period between 6000 and 3000 yr BP provides multiple opportunities for study. These opportunities include not only the *Tsuga* decline itself, as noted by Booth et al. (2012a). Additional records with comparably high temporal resolution and macrofossil sensitivity could determine whether the sequence of forest changes between 6000 and 5000 yr BP (e.g., the declines in *P. strobus*, *A. saccharum*, and *B. papyrifera*) occurred at other sites, and whether these events were synchronous. Assessment of whether the transient *Tsuga* declines 5500 and 5200 yr BP occurred at other sites within the region and in other regions can help determine causes and consequences (Booth et al., 2012a). The macrofossil declines accompanying the pollen declines at Tower Lake indicate that these events represent actual population declines, rather than passing fluctuations in pollen productivity (e.g.,

Minckley et al., 2012). Development of carefully designed site networks to identify the fundamental ecological and paleoclimatic patterns in space and time, together with hypothesis-testing to evaluate potential mechanisms and interactions, can speak directly to questions of ecological and biogeographic responses to environmental change at temporal and spatial scales that matter greatly to resource managers.

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STJ, RKB, KR, TAM, and JJA designed the project; KR, RKB, JJA, TAM, RAJ, and STJ analyzed data; STJ wrote the paper with editorial input from RKB and TAM.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.quascirev.2014.05.030>.

References

- Albert, D.A., Comer, P.J., 2008. Atlas of Early Michigan's Forests, Grasslands, and Wetlands: an Interpretation of the 1816–1856. General Land Office Surveys. Michigan State University Press, East Lansing.
- Anderson, R.S., Davis, R.B., Miller, N.G., Stuckenrath, R., 1986. History of late- and post-glacial vegetation and disturbance around Upper South Branch Pond, northern Maine. *Can. J. Bot.* 64, 1977–1986.
- Barnekow, L., Sandgren, P., 2001. Palaeoclimate and tree-line changes during the Holocene based on pollen and plant macrofossil records from six lakes at different altitudes in northern Sweden. *Rev. Palaeobot. Palynol.* 117, 109–118.
- Bartlein, P.J., Webb III, T., Fleri, E., 1984. Holocene climatic change in the northern Midwest: pollen-derived estimates. *Quat. Res.* 22, 361–374.
- Bartlein, P.J., Harrison, S.P., Brewer, S., Connor, S., Davis, B.A.S., Gajewski, K., Guiot, J., Harrison-Prentice, T.L., Henderson, A., Peyron, O., Prentice, I.C., Scholze, M., Seppä, H., Shuman, B., Sugita, S., Thompson, R.S., Viau, A.E., Williams, J., Wu, H., 2011. Pollen-based continental climate reconstructions at 6 and 21 ka: a global synthesis. *Clim. Dyn.* 37, 775–802.
- Betancourt, J.L., Van Devender, T.R., Martin, P.S. (Eds.), 1990. *Packrat Middens: the Last 40,000 Years of Biotic Change*. University of Arizona Press, Tucson.
- Binney, H.A., Willis, K.J., Edwards, M.E., Bhagwat, S.A., Anderson, P.M., Andreev, A.A., Blaauw, M., Damblon, F., Haesaerts, P., Kienast, F., Kremenetski, K.V., Krivonogov, S.K., Lozhkin, A.V., MacDonald, G.M., Novenko, P.O., Sapelko, T., Väliranta, M., Vazhenina, L., 2009. The distribution of late-Quaternary woody taxa in northern Eurasia: evidence from a new macrofossil database. *Quat. Sci. Rev.* 28, 2445–2464.
- Birks, H.H., 1973. Modern macrofossil assemblages in lake sediments in Minnesota. In: Birks, H.J.B., West, R.G. (Eds.), *Quaternary Plant Ecology*. Blackwells, Oxford, England, pp. 173–189.
- Blaauw, M., Christen, J.A., 2011. Flexible paleoclimate age–depth models using an autoregressive gamma process. *Bayesian Anal.* 6, 457–474.
- Blarquez, O., Carcaillet, C., Elzein, T.M., Roiron, P., 2012. Needle accumulation rate model-based reconstruction of palaeo-tree biomass in the western subalpine Alps. *Holocene* 22, 579–587.
- Booth, R.K., Jackson, S.T., Gray, C.E.D., 2004. Paleoecology and high-resolution paleohydrology of a kettle peatland in upper Michigan. *Quat. Res.* 61, 1–13.
- Booth, R.K., Jackson, S.T., Forman, S.L., Kutzbach, J.E., Bettis III, E.A., Kreig, J., Wright, D.K., 2005. A severe centennial-scale drought in mid-continental North America 4200 years ago and apparent global linkages. *Holocene* 15, 321–328.
- Booth, R.K., Brewer, S., Blaauw, M., Minckley, T.A., Jackson, S.T., 2012a. Decomposing the mid-Holocene *Tsuga* decline in eastern North America. *Ecology* 93, 1841–1852.
- Booth, R.K., Jackson, S.T., Sousa, V.A., Sullivan, M.E., Minckley, T.A., Clifford, M., 2012b. Multidecadal drought and amplified moisture variability drove rapid forest community change in a humid region. *Ecology* 93, 219–226.

- Bradshaw, R.H.W., Webb III, T., 1985. Relationships between contemporary pollen and vegetation data from Wisconsin and Michigan, USA. *Ecology* 66, 721–737.
- Brewer, S., Jackson, S.T., Williams, J.W., 2012. Paleoecoinformatics: applying geo-historical data to ecological questions. *Trends Ecol. Evol.* 27, 104–112.
- Brubaker, L.B., 1975. Postglacial forest patterns associated with till and outwash in north-central Upper Michigan. *Quat. Res.* 5, 499–527.
- Davis, M., Douglas, C., Calcote, R., Cole, K.L., Winkler, M.G., Flakne, R., 2000. Holocene climate in the western Great Lakes National Parks and Lakeshores: implications for future climate change. *Conserv. Biol.* 14, 968–983.
- Davis, M.B., Schwartz, M.W., Woods, K., 1991. Detecting a species limit from pollen in sediments. *J. Biogeogr.* 18, 653–668.
- Drake, H., Burrows, C.J., 1980. The influx of potential macrofossils into Lady Lake, north Westland, New Zealand. *N. Z. J. Bot.* 18, 257–274.
- Dunwiddie, P.W., 1986. A 6000-year record of forest history on Mount Rainier, Washington. *Ecology* 67, 58–68.
- Dunwiddie, P.W., 1987. Macrofossil and pollen representation of coniferous trees in modern sediments from Washington. *Ecology* 68, 1–11.
- Ewing, H.A., 2002. The influence of substrate on vegetation history and ecosystem development. *Ecology* 83, 2766–2781.
- Fægri, K., Iversen, J., 1975. *Text-Book of Pollen Analysis*. Third Revised Edition. Munksgaard, Copenhagen, 295 p.
- Fall, P.L., 1992. Spatial patterns of atmospheric pollen dispersal in the Colorado Rocky Mountains, USA. *Rev. Palaeobot. Palynol.* 74, 293–313.
- Farrand, W.R., Bell, D.L., 1982. *Quaternary Geology of Northern Michigan* (Map). Michigan Department of Natural Resources, Geological Survey Division, Lansing.
- Finsinger, W., Tinner, W., 2007. Pollen and plant macrofossils at Lac de Fully (2135 m a.s.l.): Holocene forest dynamics on a highland plateau in the Valais, Switzerland. *Holocene* 17, 1119–1127.
- Ford, M.S., 1990. A 10 000-yr history of natural ecosystem acidification. *Ecol. Monogr.* 60, 57–89.
- Foster, D.R., Oswald, W.W., Faison, E.K., Doughty, E.D., Hansen, B.C.S., 2006. A climatic driver for abrupt mid-Holocene vegetation dynamics and the hemlock decline in New England. *Ecology* 87, 2959–2966.
- Frelich, L.E., Calcote, R.R., Davis, M.B., Pastor, J., 1993. Patch formation and maintenance in an old-growth hemlock-hardwood forest. *Ecology* 74, 513–527.
- Fuller, J.L., 1998. Ecological impact of the mid-Holocene hemlock decline in southern Ontario, Canada. *Ecology* 79, 2337–2351.
- Gavin, D.G., Brubaker, L.B., Greenwald, D.N., 2013. Postglacial climate and fire-mediated vegetation change on the western Olympic Peninsula, Washington (USA). *Ecology* 83, 471–489.
- Gill, J.L., Williams, J.W., Jackson, S.T., Lininger, K.B., Robinson, G.S., 2009. Pleistocene megafaunal collapse, novel plant communities, and enhanced fire regimes in North America. *Science* 326, 1100–1103.
- Gill, J.L., Williams, J.W., Jackson, S.T., Donnelly, J.P., Schellinger, G.C., 2012. Climatic and megaherbivory controls on late-glacial vegetation dynamics: a new, high-resolution, multi-proxy record from Silver Lake, Ohio. *Quat. Sci. Rev.* 34, 66–80.
- Goring, S.J., Williams, J.W., Blois, J.L., Jackson, S.T., Paciorek, C.J., Booth, R.K., Marlon, J.R., Blaauw, M., Christen, J.A., 2012. Deposition times in the northeastern United States during the Holocene: establishing valid priors for Bayesian age models. *Quat. Sci. Rev.* 48, 54–60.
- Gray, S.T., Betancourt, J.L., Jackson, S.T., Eddy, R.G., 2006. Role of multidecadal climatic variability in a range extension of pinyon pine. *Ecology* 87, 1124–1130.
- Grimm, E.C., 1987. CONISS: a FORTRAN 77 program for stratigraphically constrained cluster analysis by the method of incremental sum of squares. *Comput. Geosci.* 13, 13–35.
- Harlow, W.M., 1947. *The Identification of the Pines of the United States*, Native and Introduced, by Needle Structure. Technical Publication No. 32. New York State College of Forestry, Syracuse, New York.
- He, H.S., Mladenoff, D.J., Sickley, T.A., Guntenspergen, G.G., 2000. GIS interpolation of witness tree records (1839–1866) for northern Wisconsin at multiple scales. *J. Biogeogr.* 27, 1031–1042.
- Hickler, T., Smith, B., Sykes, M.T., Davis, M.B., Sugita, S., Walker, K., 2004. Using a generalized vegetation model to simulate vegetation dynamics in northeastern USA. *Ecology* 85, 519–530.
- Hotchkiss, S.C., Calcote, R., Lynch, E.A., 2007. Response of vegetation and fire to Little Ice Age climate change: regional continuity and landscape heterogeneity. *Landsc. Ecol.* 22, 25–41.
- Jackson, S.T., 1989. *Postglacial Vegetational Changes Along an Elevational Gradient in the Adirondack Mountains* (New York): a Study of Plant Macrofossils. New York State Museum and Science Service Bulletin 465, Albany.
- Jackson, S.T., 1990. Pollen source area and representation in small lakes of the northeastern United States. *Rev. Palaeobot. Palynol.* 63, 53–76.
- Jackson, S.T., 1991. Pollen representation of vegetational patterns along an elevational gradient. *J. Veg. Sci.* 2, 613–624.
- Jackson, S.T., 1994. Pollen and spores in Quaternary lake sediments as sensors of vegetation composition: theoretical models and empirical evidence. In: Traverse, A. (Ed.), *Sedimentation of Organic Particles*. Cambridge University Press, Cambridge, pp. 253–286.
- Jackson, S.T., 1999. Techniques for analysing unconsolidated lake sediments. In: Jones, T., Rowe, N. (Eds.), *Fossil Plants and Spores: Modern Techniques*. Geological Society, London, pp. 274–278.
- Jackson, S.T., 2012. Representation of flora and vegetation in Quaternary fossil assemblages: known and unknown knowns and unknowns. *Quat. Sci. Rev.* 49, 1–15.
- Jackson, S.T., Betancourt, J.L., Booth, R.K., Gray, S.T., 2009. Ecology and the ratchet of events: climate variability, niche dimensions, and species distributions. *Proc. Natl. Acad. Sci.* 106, 19685–19692.
- Jackson, S.T., Blois, J.L., 2014. The ecological community in a changing environment: perspectives from the Quaternary fossil record. *Proc. Natl. Acad. Sci.* (in revision).
- Jackson, S.T., Booth, R.K., 2007. Validation of pollen studies. In: Elias, S.A. (Ed.), *Encyclopaedia of Quaternary Sciences*. Elsevier Scientific Publishing, Inc., Amsterdam, pp. 2413–2422.
- Jackson, S.T., Lyford, M.E., 1999. Pollen dispersal models in Quaternary plant ecology: assumptions, parameters, and prescriptions. *Bot. Rev.* 65, 39–75.
- Jackson, S.T., Overpeck, J.T., Webb III, T., Keatts, S.E., Anderson, K.H., 1997. Mapped plant macrofossil and pollen records of Late Quaternary vegetation change in eastern North America. *Quat. Sci. Rev.* 16, 1–70.
- Jackson, S.T., Smith, S.J., 1994. Pollen dispersal and representation on an isolated, forested plateau. *New Phytol.* 128, 181–193.
- Jackson, S.T., Webb, R.S., Anderson, K.H., Overpeck, J.T., Webb III, T., Williams, J., Hansen, B.C.S., 2000. Vegetation and environment in unglaciated eastern North America during the Last Glacial maximum. *Quat. Sci. Rev.* 19, 489–508.
- Jackson, S.T., Whitehead, D.R., 1991. Holocene vegetation patterns in the Adirondack Mountains. *Ecology* 72, 641–653.
- Jacobson Jr., G.L., Bradshaw, R.H.W., 1981. The selection of sites for paleovegetational studies. *Quat. Res.* 16, 80–96.
- Lesser, M.R., Jackson, S.T., 2012. Making a stand: five centuries of population growth in colonizing stands of *Pinus ponderosa*. *Ecology* 93, 1071–1081.
- Liu, Y., Brewer, S., Booth, R.K., Minckley, T.A., Jackson, S.T., 2012. The effect of temporal density of pollen sampling on age-determination of the mid-Holocene *Tsuga canadensis* (hemlock) decline in eastern North America. *Quat. Sci. Rev.* 45, 54–59.
- Lotter, A.F., 1999. Late-glacial and Holocene vegetation history and dynamics as shown by pollen and plant macrofossil analyses in annually laminated sediments from Soppensee, central Switzerland. *Veg. Hist. Archaeobot.* 8, 165–184.
- MacDonald, G.M., 1993. Fossil pollen analysis and the reconstruction of plant invasions. *Adv. Ecol. Res.* 24, 67–110.
- Marsicek, J.P., Shuman, B., Brewer, S., Foster, D.R., Oswald, W.W., 2013. Moisture and temperature changes associated with the mid-Holocene *Tsuga* decline in the northeastern United States. *Quat. Sci. Rev.* 80, 129–142.
- McLachlan, J.S., Clark, J.S., 2004. Reconstructing historical ranges with fossil data at continental scales. *For. Ecol. Manag.* 197, 139–147.
- Miller, P.A., Giesecke, T., Hickler, T., Bradshaw, R.H.W., Smith, B., Seppä, H., Valdes, P.J., Sykes, M.T., 2008. Exploring climatic and biotic controls on Holocene vegetation change in Fennoscandia. *J. Ecol.* 96, 247–259.
- Minckley, T.A., Booth, R.K., Jackson, S.T., 2012. Response of regional pollen assemblages to late Holocene drought events in the Upper Midwest, USA. *Holocene* 22, 531–539.
- National Research Council, Committee on the Geological Record of Biosphere Dynamics, 2005. *The Geologic Record of Ecological Dynamics: Understanding the Biotic Consequences of Global Change*. National Academies Press, Washington, DC, 200 pages.
- Oswald, W.W., Foster, D.R., 2011. Middle-Holocene dynamics of *Tsuga canadensis* (eastern hemlock) in northern New England, USA. *Holocene* 22, 71–78.
- Pederson, N., Dyer, J.M., McEwan, R.W., Hessel, A.E., Mock, C.J., Orwig, D.A., Rieder, H.E., Cook, B.I., 2014. The legacy of episodic climatic events in shaping temperate, broadleaf forests. *Ecol. Monogr.* <http://dx.doi.org/10.1890/13-1025.1>.
- Pellatt, M.G., Mathewes, R.W., 1994. Paleoeecology of postglacial tree line fluctuations on the Queen Charlotte Islands, Canada. *Écoscience* 1, 71–81.
- Pellatt, M.G., Mathewes, R.W., 1997. Holocene tree line and climate change on the Queen Charlotte Islands, Canada. *Quat. Res.* 48, 88–99.
- Peters, D.P.C., Pielke Sr., R.A., Bestelmayer, B.T., Allen, C.D., Munson-McGee, S., Havstad, K.M., 2004. Cross-scale interactions, nonlinearities, and forecasting catastrophic events. *Proc. Natl. Acad. Sci.* 101, 15130–15135.
- Peters, D.P.C., Bestelmayer, B.T., Turner, M.G., 2007. Cross-scale interactions and changing pattern–process relationships: consequences for system dynamics. *Ecosystems* 10, 790–796.
- Prentice, I.C., 1985. Pollen representation, source area, and basin size: toward a unified theory of pollen analysis. *Quat. Res.* 23, 76–86.
- Reimer, P.J., Baillie, M.G.L., Bard, E., Bayliss, A., Beck, J.W., Blackwell, P.G., Bronk Ramsey, C., Buck, C.E., Burr, G.S., Edwards, R.L., Friedrich, M., Grootes, P.M., Guilderson, T.P., Hajdas, I., Heaton, T.J., Hogg, A.G., Hughen, K.A., Kaiser, K.F., Kromer, B., McCormac, F.G., Manning, S.W., Reimer, R.W., Richards, D.A., Southon, J.R., Talamo, S., Turney, C.S.M., van der Plicht, J., Weyhenmeyer, C.E., 2009. IntCal09 and Marine09 radiocarbon age calibration curves, 0–50,000 cal BP. *Radiocarbon* 51, 1111–1150.
- Schulte, L.R., Mladenoff, D.J., Crow, T.R., Merrick, L.C., Cleland, D.T., 2007. Homogenization of northern U.S. Great Lakes forests due to land use. *Landsc. Ecol.* 22, 1089–1103.
- Shuman, B., 2012. Patterns, processes, and impacts of abrupt climate change in a warm world: the past 11,700 years. *WIREs Clim. Change* 3, 19–43.
- Shuman, B., Newby, P., Huang, Y., Webb III, T., 2004. Evidence for the close climatic control of New England vegetation history. *Ecology* 85, 1297–1310.

- Shuman, B.N., Newby, P., Donnelly, J.P., 2009a. Abrupt climate change as an important agent of ecological change in the Northeast U.S. through the past 15,000 years. *Quat. Sci. Rev.* 28, 1693–1709.
- Shuman, B., Henderson, A.K., Plank, C., Stefanova, I., Ziegler, S.S., 2009b. Woodland-to-forest transition during prolonged drought in Minnesota after ca. AD 1300. *Ecology* 90, 2792–2807.
- Singer, D.K., Jackson, S.T., Madsen, B.J., Wilcox, D.A., 1996. Differentiating climatic and successional influences on long-term development of a marsh. *Ecology* 77, 1765–1778.
- Soepboer, W., Sugita, S., Lotter, A.F., 2010. Regional vegetation-cover changes on the Swiss Plateau during the past two millennia: a pollen-based reconstruction using the REVEALS model. *Quat. Sci. Rev.* 29, 472–483.
- Spear, R.W., 1989. Late-Quaternary history of high-elevation vegetation in the White Mountains of New Hampshire. *Ecol. Monogr.* 59, 125–151.
- Spear, R.W., Davis, M.B., Shane, L.C.K., 1994. Late Quaternary history of low- and mid-elevation vegetation in the White Mountains of New Hampshire. *Ecol. Monogr.* 64, 85–109.
- Steenstrup, J.J.S., 1841. Kongelige Danske Videnskabernes Selskabs Afhandlinger. Geognostik-Geologisk Undersøgelse av Skovmosserne Vidnesam- og Lille-mose i det nordlige Sjælland, ledsaget af sammenlignende Bemærkninger, hentede fra Danmarks Skov-, Kjør og Lyngmoser i Almindelighed, vol. 9, pp. 17–120.
- Sugita, S., 1994. Pollen representation of vegetation in Quaternary sediments: theory and method in patchy vegetation. *J. Ecol.* 82, 881–897.
- Sugita, S., 2007a. Theory of quantitative reconstruction of vegetation I: pollen from large sites REVEALS regional vegetation composition. *Holocene* 17, 229–241.
- Sugita, S., 2007b. Theory of quantitative reconstruction of vegetation II: all you need is LOVE. *Holocene* 17, 243–257.
- Sugita, S., Parshall, T., Calcote, R., Walker, K., 2010. Testing the Landscape Reconstruction Algorithm for spatially explicit reconstruction of vegetation in northern Michigan and Wisconsin. *Quat. Res.* 74, 289–300.
- Swetnam, T.W., Betancourt, J.L., 1998. Mesoscale disturbance and ecological response to decadal climatic variability in the American Southwest. *J. Clim.* 11, 3128–3147.
- Tinner, W., Theurillat, J.-P., 2003. Uppermost limit, extent, and fluctuations of the timberline and treeline ecocline in the Swiss Central Alps during the past 11,500 years. *Arct. Antarct. Alp. Res.* 35, 158–169.
- Tweiten, M.A., Hotchkiss, S.C., Booth, R.K., Calcote, R.R., Lynch, E.A., 2009. The response of a jack pine forest to late-Holocene climate variability in north-western Wisconsin. *Holocene* 19, 1049–1061.
- Vaupell, C., 1851. De Nordsjællandske Skovmoser. En Botanisk-Mikroskopisk Undersøgelse af de Plantedele, som danne Törven, og af de Levninger af Fortidens Skove, der ere Bevarede i Nogle Nordsjællandske Skovmoser. C.A. Reitzel, Copenhagen, 56 p.
- Von Post, L., 1916 (1967). Forest tree pollen in south Sweden bog deposits. *Pollen Spores* 9, 375–401.
- Wainman, N., Mathewes, R.W., 1987. Forest history of the last 12 000 years based on plant macrofossil analysis of sediment from Marion Lake, southwestern British Columbia. *Can. J. Bot.* 65, 2179–2187.
- Wainman, N., Mathewes, R.W., 1990. Distribution of plant macroremains in surface sediments of Marion Lake, southwestern British Columbia. *Can. J. Bot.* 68, 364–373.
- Watts, W.A., Winter, T.C., 1966. Plant macrofossils from Kirchner Marsh, Minnesota—a paleoecological study. *Geol. Soc. Am. Bull.* 77, 1339–1360.
- Webb III, T., 1986. Is vegetation in equilibrium with climate? How to interpret late-Quaternary pollen data. *Vegetatio* 67, 75–91.
- Webb III, T., 1987. The appearance and disappearance of major vegetational assemblages: long-term vegetational dynamics in eastern North America. *Vegetatio* 69, 177–187.
- Wick, L., van Leeuwen, J.F.N., van der Knapp, W.O., Lotter, A.F., 2003. Holocene vegetation development in the catchment of Sägistalsee (1935 m asl), a small lake in the Swiss Alps. *J. Paleolimnol.* 30, 261–272.
- Williams, J.W., Shuman, B., Webb III, T., Bartlein, P.J., Leduc, P.L., 2004. Late-Quaternary vegetation dynamics in North America: scaling from taxa to biomes. *Ecol. Monogr.* 74, 309–334.
- Williams, J.W., Shuman, B., Bartlein, P.J., Diffenbaugh, N.S., Webb III, T., 2010. Rapid, time-transgressive, and variable responses to early Holocene midcontinental drying in North America. *Geology* 38, 135–138.
- Woods, K.D., 2000a. Dynamics in late-successional hemlock-hardwoods forests over three decades. *Ecology* 81, 110–126.
- Woods, K.D., 2000b. Long-term change and spatial pattern in a late-successional hemlock-northern hardwood forest. *J. Ecol.* 88, 267–282.
- Yu, Z., 1997. Late Quaternary paleoecology of *Thuja* and *Juniperus* (Cupressaceae) at Craford Lake, Ontario, Canada: pollen, stomata and macrofossils. *Rev. Palaeobot. Palynol.* 96, 241–354.